Amazon deforestation alert
Amazon deforestation is about to cross a tipping point after which it will undergo irreversible changes. **http://dgit.in/AmazonThreat**

SpaceX plans to expand to LA Port
The aeronautics giant SpaceX is looking to expand its projects in the city of Los Angles for working on rockets for Mars **http://dgit.in/SpaceXLA**

Bellatrix's MET thruster design is supposed to be highly scalable starting from Nano satellites to micro satellites. According to them, it should be able to generate between a few micronewton of thrust to 200 millinewton of thrust. This system can be used for both orbit raising and orbital station-keeping. Another advantage is the higher thrust-to-power ratio achieved in this system compared to others. When the microwave energy is injected into the cavity, the plasma generated doesn't touch the walls of the thruster. The resonant cavity design ensures the plasma freely floats inside without touching the walls. The same applies to the nozzle. Essentially, the thruster gets higher durability.

Microwave electrothermal thruster model

PROTOTYPING PHASE

As mentioned earlier, Bellatrix hasn't invented a new propulsion system. They have simply worked out all the specifications and created a physical system that proves the technology works. Since these propulsion systems aren't bulky, the prototypes were built to actual scale. Yashas recalls that their prototype would simply lay idle and there wouldn't be any plasma formation. At its early stages, the same procedure was being used but it would fail. Different configurations let to different problems such as the water corroding the thruster. Sometimes the plasma wouldn't be stable enough and not work for a long period.

All these experiments were conducted inside their own lab. Before moving in, the prototyping was conducted with support from Hindustan College of Engineering and Technology, Coimbatore. They built their own vacuum chamber facility to test the prototype thruster. After those numerous trials and changes in configurations, the team was able to successfully generate thrust using water in 2015. The water didn't need special storage conditions and it was in room temperature as well. Yashas claims that no one else in the world was able to achieve thrust using water as a propellant before using their type of propulsion system.

The team hasn't finished building and testing all the components of the engine yet. So, there's still time until they can test the MET in its full glory. However, simulations have been successfully carried out for the entire system.

WORKING WITH ISRO

On a global scale, the Indian Space Research Organisation (ISRO) has already impressed everyone with its potential in space research. Hence, it was only natural for the team from Bellatrix Aerospace to seek opportunities at ISRO. As college students, they weren't able to figure out the correct approach to get in touch with ISRO. Fortunately, former ISRO chairman, Mr. A. S. Kiran Kumar, was the guest of honour at their convocation ceremony. They got the opportunity to present their work to him and were told to expect a call later. Mr. A S Kiran Kumar escalated the project at ISRO and they were contacted after a few months. Both the co-founders were delighted and surprised on being taken seriously by the government's space body so early into their journey. And ISRO has been actively following up with their project. After their collaboration with ISRO kicked off, Yashas attributed a lot of support to Dr. P. S. Goel and Dr. B. N. Suresh, especially suggestions for design improvements and also guiding other projects.

Chetak - Two-stage launch vehicle with a reusable first-stage

Already having delivered a working prototype, the company was able to bag a contract with ISRO. The space agency wanted them to build the microwave electrothermal thruster according to specific configurations and requirements of their satellites. Generally, any company that receives a tender from ISRO has to pay a certain security deposit and provide a performance bank guarantee, among other clauses. As a startup, Bellatrix didn't really possess the required funds and hence, they couldn't think of going ahead with the offer. Fortunately, according to Yashas, ISRO had an internal policy change and they accepted Bellatrix's request of waiving off the clauses. The company got the contract. Yashas mentioned that ISRO has provided incredible support to the company in terms of technical knowledge. Every time they visit the space center to test their thruster, they are always learning new things from the bright minds working there.

Apart from ISRO, Bellatrix also received an order from the Bengaluru division of Liquid Propulsion System Centre (LPSC). Collaboration with LPSC gave them knowledge on how ISRO validates all the technologies. Bellatrix wants to learn and incorporate a similar level of validation techniques into their own projects.

FUTURE GOALS

There are quite a few things that Bellatrix Aerospace is working on for the future. In launch vehicles, one of their focus areas, they've developed a couple of designs, Chetak and Garuda, that can carry a payload of 500kg and 1010kg respectively. They're also working on a Green Monopropellant Thruster (HAN blend), which is a cleaner alternative to Monopropellant Hydrazine Thrusters, a toxic chemical used in chemical propulsion systems. The company is also developing Hall Thrusters, a type of ion thruster, with specific improvements to extend the expected life of these kind of thrusters. d